When Overall AUC Hides Miscalibration: Evaluating Knowledge Tracing on Rarely Practiced Knowledge Components

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Abstract*—Knowledge-tracing models are usually ranked by overall AUC, which can hide calibration differences among knowledge components that learners rarely practice. We lock one protocol on ASSISTments 2012: the validation-AUC checkpoint, test windows of at most 200, 15 equal-width bins, and four unique learner partitions. Deep knowledge tracing had mean ECE 0.0982 on sparse components and 0.0089 on dense components (ΔECE = 0.0893). Its fold-0 size-matched gap was 0.0351 (learner-cluster 95% interval 0.0170 to 0.0821). The four-partition bias reference ΔECE* was −0.0081 and is not a significance test. A causally masked Transformer under the same protocol had a fold-0 matched interval of 0.0233 to 0.0878 and a four-partition ΔECE* of 0.0068. Junyi Academy has no sparse group; its post-hoc medium-versus-dense ΔECE* is −0.0032. XES3G5M is a descriptive audit only, after a failed leakage check and no corrected retraining. Groups use training counts only, and empty test windows are dropped. The four-partition SD is descriptive, because the partitions are not independent samples. Unmasked Transformer scores are excluded. Overall AUC can hide frequency-stratified calibration differences, but their size and presence depend on the protocol, the checkpoint, the sample size, the model, and the dataset.

***Keywords*—knowledge tracing, calibration, knowledge components, adaptive learning, learning analytics, evaluation protocol**

I. INTRODUCTION

Adaptive practice platforms and intelligent tutoring systems use knowledge tracing (KT) to estimate how well a learner has mastered a skill and to decide whether to give more practice, assign remediation, or move on [1]–[3]. Such decisions often rest on the predicted probability p that the learner will answer the next item on the skill correctly: once p reaches a threshold τ, further practice on that skill is withheld. A rule of this kind takes p at face value. A prediction of 0.7 should mean that about seven in ten comparable responses are correct; in other words, the model must be well calibrated.

KT models, however, are usually compared by the area under the receiver operating characteristic curve (AUC) and accuracy computed over all test responses [4], [5]. AUC measures ranking, that is, whether correct responses tend to receive higher probabilities than incorrect ones; calibration is a separate property. The difference matters most for knowledge components (KCs) that learners rarely practice. In the training data used here, 18.9% of the KCs in ASSISTments 2012 and 21.5% of the XES3G5M KCs present in training had fewer than 100 training rows (Section IV.C), whereas frequently practiced KCs supplied most of the rows. Pooled metrics therefore mainly reflect frequently practiced KCs, and a model can rank responses well overall while assigning overconfident probabilities to rarely practiced KCs. Under a fixed threshold, such overconfidence appears as an advance decision followed by an incorrect next response; we call the proportion of these decisions the false-advance rate (FAR).

Earlier work has examined parts of this problem. Benchmark studies have standardized KT evaluation and documented label leakage [5], [6], and calibration, uncertainty, cold start, and mastery thresholds have been studied in their own right (Section II). This study combines these elements into one evaluation protocol for standard KT models; it does not propose a new model.

The study addresses three research questions. Only ASSISTments 2012, under protocol P0-BV200-15EW, is used for the primary answers. Junyi Academy is a post-hoc check. XES3G5M contributes distribution and estimability, not model scores.

RQ1: Under the locked protocol, how does ECE differ between sparse and dense knowledge components on ASSISTments 2012, once the bias reference and a size-matched comparison are reported separately?

RQ2: When is a sparse-versus-dense contrast estimable, and when must model-based results be withheld?

RQ3: On ASSISTments 2012, how does the DKT contrast change when the checkpoint and the test window are varied one at a time?

For RQ1, DKT and a causally masked Transformer are the primary models. Across four unique partitions their sparse ECE exceeds their dense ECE. The size-matched gap is positive, and the fold-0 interval for that gap excludes 0. ΔECE* does not stay positive on every partition, so the bias reference and the matched interval are not the same statement. Section IV gives the partition means. For RQ2, Junyi Academy has no sparse group. XES3G5M has a low-frequency tail in its KC-level file, but check L8 failed and no corrected model was trained, so it is a descriptive audit only. For RQ3, the DKT grid in Table A2 separates the final checkpoint from the validation-AUC checkpoint and the full test history from windows of 200. Changing both at once is not attributed to either factor alone.

The paper makes three contributions. First, it specifies a protocol that groups knowledge components by training frequency, labels sample support, defines ΔECE* as a bias reference rather than a test, and states eight leakage checks. Second, it reports ASSISTments DKT and T-KT masked under that protocol, with the raw contrast, the bias reference, and the size-matched gap kept apart. Third, it states the limits of the evidence: Junyi Academy is post-hoc and has no sparse group, XES3G5M is a descriptive audit only, and a change of both checkpoint and test window is not a single-cause result.

Unmasked T-KT is not used as a finding.

II. LITERATURE REVIEW

A. Knowledge tracing models and benchmarks

KT models a learner's changing proficiency from past interactions in order to predict the next response [3]. Bayesian knowledge tracing (BKT) tracks a latent mastery state for each skill [1]. Item response theory (IRT), in its one-parameter Rasch form [7], relates the probability of a correct answer to learner ability and item difficulty. DKT replaced hand-specified learning dynamics with a recurrent network and established next-response prediction as the standard task [2]. Attention-based successors, such as context-aware attentive knowledge tracing [8] and simpleKT [4], keep that task but replace recurrence with self-attention. The best model depends on the data: in a comparison on nine real-world datasets, logistic regression with a suitable feature set led on datasets of moderate size, whereas DKT led on the largest ones [9]. Uncertainty-aware models such as UKT represent knowledge states as distributions [10].

B. Evaluation rigor and leakage

Benchmark libraries such as pyKT have standardized preprocessing and model comparison and have shown that evaluation choices can distort reported discrimination [5]. One such choice concerns questions tagged with several KCs. When such a question is expanded into one row per KC and each row is scored separately, a model can use the answer to the same question from the previous row. In pyKT's experiments, this raised AUC by 8.38 and 13.09 percentage points on average on ASSISTments 2009 and Algebra 2005, respectively [5]. Badran and Preisach analyze the same form of label leakage [6], and our protocol defines this route as check L8. The protocol applies the same discipline to frequency groups and calibration: each KC's group and descriptors are computed from training data only, and no cut point or threshold is tuned on test data. Graph-based KT [11] and contrastive learning for KT [12] form related research lines that focus on new model architectures.

C. Four meanings of sparsity

The word "sparse" has at least four meanings in KT research. (a) Sparse attention: sparseKT applies *k*-sparse attention so that only a subset of past interactions receives nonzero weight [13]. (b) Sparse KC frequency: a KC with few observations in the training data. (c) New-student cold start: prediction for learners with little or no history at test time [14]. (d) Concept-level cold start: a KC that never appears in the training data, whether or not the learner is new. These problems are distinct: sparse attention can be applied to a frequently practiced skill, and a learner seen in training can meet a KC absent from it. This study addresses meanings (b) and (d) through training-frequency groups that include a strict cold-start group ($f_{\text{train}} = 0$).

D. Calibration and educational decisions

When predicted probabilities drive educational decisions, ranking alone is an insufficient evaluation [15]. Calibration asks whether a predicted probability matches the observed frequency of correct responses. ECE summarizes the mismatch over probability bins [16], and modern neural networks can be poorly calibrated even when they discriminate well [17]. The Brier score [18], Murphy's decomposition into uncertainty, resolution, and reliability [19], and the calibration–refinement view [20] separate miscalibration from the difficulty of the task, while reliability diagrams display the same comparison graphically.

Within KT, post-hoc per-item bias correction has been used to recover discrimination in a trained model [21], and selective prediction defers the least certain predictions to a teacher [22]. Mastery-learning research has examined false positives at the mastery threshold [23] and predictive performance when a student starts a skill [24]. Our threshold analysis complements selective prediction: it does not abstain, but measures how often the advances made under a fixed threshold are followed by an incorrect response, separately for each frequency group.

III. MATERIALS AND METHODS

Fig. 1 outlines the evaluation workflow; the leakage checks attached to its steps are defined in Section III.F.

A. Datasets

We used three de-identified public logs mapped to a common interaction schema: ASSISTments 2012 [25], Junyi Academy [26], and XES3G5M [27]. Preprocessing removed rows with missing learner, KC, or label identifiers, binarized correctness, parsed timestamps, and kept learners with at least two interactions. Table 1 reports the counts after these filters, which therefore differ from those of the raw public files. The processed logs contain 2,657,490 (ASSISTments 2012), 16,215,567 (Junyi Academy), and 6,413,353 (XES3G5M) rows.

Table 1. Cohort statistics after preprocessing (learner-based split; test events from fold 0). For XES3G5M, interactions and test events are KC-level rows (Section III.A).

Dataset	Learners	KCs	Interactions	Test events
ASSISTments 2012	27,806	265	2.66M	534,150
Junyi Academy	71,014	1,326	16.2M	3,269,022
XES3G5M	18,066	865	6.41M	1,282,422

ASSISTments 2012 is the 2012–2013 public release with affect [25], which is also supported by the pyKT toolkit [28]; each retained row carries one skill_id. Junyi Academy is the Kaggle Online Learning Activity dataset [26], which differs from the 2015 Junyi release. We used its ucid field, the content (exercise) identifier that the practice log assigns to each item, as the KC; it is an operational identifier rather than a curated skill tag. Practice volume per exercise is high: across the whole log, the 1,326 exercises received about 12,200 responses each on average. In learner-based fold 0, every exercise had at least 100 training responses (Fig. 2), and the sparse group was empty in all four learner-based partitions, so Junyi Academy has no sparse group under the learner-based split.

XES3G5M was loaded from the authors' KC-level release: the training/validation file and the test file were concatenated and flattened to one row per valid sequence position, with padding, invalid labels, and rows with a negative KC

identifier removed. The dataset paper reports 18,066 students, 7,652 questions, 865 KCs, and 5,549,635 question-level interactions [27]. Our file has 6,413,353 rows because a question tagged with several KCs is expanded into one row per KC; the number of distinct items remains 7,652, and all XES3G5M counts, including the 1,282,422 test events of fold 0 (Table 1), are KC-level rows. If the model input for one row contains the outcome recorded at another row of the same response, the target label is visible to the model [5], [6]. Our audit of this route (check L8, Section III.F) in the code used for the XES3G5M runs found that the KC rows of a response stay adjacent and that both training and scoring condition each prediction on the KC–response tokens of earlier rows. For the second and later KC rows of a multi-KC response (863,807 rows by the release's repeat flags, 13.5% of the file), the input therefore contains the outcome being predicted, in training and at test time, so L8 fails. Because removing this route requires retraining, which this study did not do, we report no model-based results for XES3G5M; the dataset still contributes the frequency distribution and estimability conditions (Fig. 2 and Section IV.C), which do not depend on predicted values. In ASSISTments 2012 and Junyi Academy, each released row records one response with one KC identifier and no expansion step was applied, so the route does not arise.

B. Splits and partitions

The learner-based split is the primary design. For each partition, learners were shuffled with the partition seed, and 20% were held out for testing, 10% for validation, and the rest for training, so the three sets share no learners. A temporal split served as a complementary check: all interactions were sorted by timestamp, and the earliest 70%, the next 10%, and the latest 20% formed the training, validation, and test sets. It was run once (seed 42), and this paper reports from it only the temporal distributions in Fig. 2; all performance results come from the learner-based split. In both designs, KC frequencies, and hence group membership, were computed from the training set only.

Deep models were trained with seeds 42, 2024, 2025, 2026, and 2027 (folds 0–4). On every dataset, seeds 2025 and 2026 share one learner partition (fold 2 = fold 3), so the five runs cover four distinct partitions, and the two runs on the shared partition were averaged before averaging across partitions. The Junyi Academy runs of the Transformer model (T-KT, Section III.C) were aligned differently: seeds 42 and 2024 were two initializations on fold 0, averaged first, and seeds 2025, 2026, and 2027 were run on folds 1, 2, and 4, so these labels are initialization seeds rather than split seeds. Unless stated otherwise, every mean ± standard deviation reported here is therefore an equal-weight summary over four learner-based partitions.

C. Models

Table 2 lists the runs that supply the primary and sensitivity scores. DKT is an LSTM over KC–response tokens and pads with index 0. T-KT masked is a two-layer Transformer over the same tokens, with a causal mask and padding id 2n. Both were trained on ASSISTments 2012 for at most 50 epochs, with patience 8, and the saved checkpoint is the one with the highest validation AUC. Test sequences are scored in non-overlapping windows of at most 200. simpleKT is the pyKT implementation and is a sensitivity analysis, not a primary model. The one-parameter IRT fit returns the training base rate for unseen learners, so its AUC is 0.5000 by construction and it is not a ranking model. BKT was not scored.

The primary checkpoint is the validation-AUC checkpoint, not the final epoch. The final epoch and the full test history remain in the sensitivity grid (Table A2). That grid was pre-specified for ASSISTments DKT. Junyi DKT uses the same checkpoint and window rules, but it was not in the pre-specified plan and is not called pre-specified. The original weights behind the archived final-checkpoint pairs were not recovered. Those pairs are not results of this protocol.

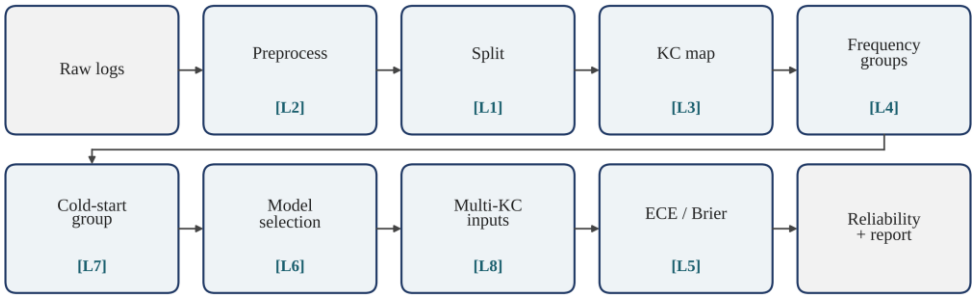


Fig. 1. Evaluation workflow. Bracketed tags give the leakage checks of Table 3 (L1–L7 numbered as in the archived audit records); they do not follow the left-to-right order. L8 applies only to datasets in which one response carries several KCs.

Table 2. Configuration of the scored neural models. Primary rows are ASSISTments DKT and T-KT masked under P0-BV200-15EW. simpleKT is sensitivity. No missing setting was filled in. The base-rate reference is not a neural model and is omitted here.

Setting	DKT	T-KT masked	simpleKT
Mask and input	LSTM; KC–response token; pad 0	Causal mask; pad id 2n; same tokens	pyKT causal attention; items and KCs
Training	ASSISTments; ≤ 50 epochs; patience 8	ASSISTments; ≤ 50 epochs; patience 8	ASSISTments; pyKT; sensitivity
Checkpoint rule	Maximum validation AUC	Maximum validation AUC	Maximum validation AUC
Test history	Non-overlapping windows ≤ 200	Non-overlapping windows ≤ 200	Windows ≤ 200

Setting	DKT	T-KT masked	simpleKT
Splits	Learner-disjoint; folds 0–4; fold 2 = fold 3	Same partitions	Same partitions
Provenance	Code 4ff6683; prediction exports	Code 4ff6683; mask tests passed	Post-hoc; not a primary model
Role	Primary	Primary	Sensitivity
Protocol	P0-BV200-15EW	P0-BV200-15EW	Same scoring rules; post-hoc
Not in this table	Unmasked T-KT is invalid	Junyi T-KT was not trained	XES models were not trained

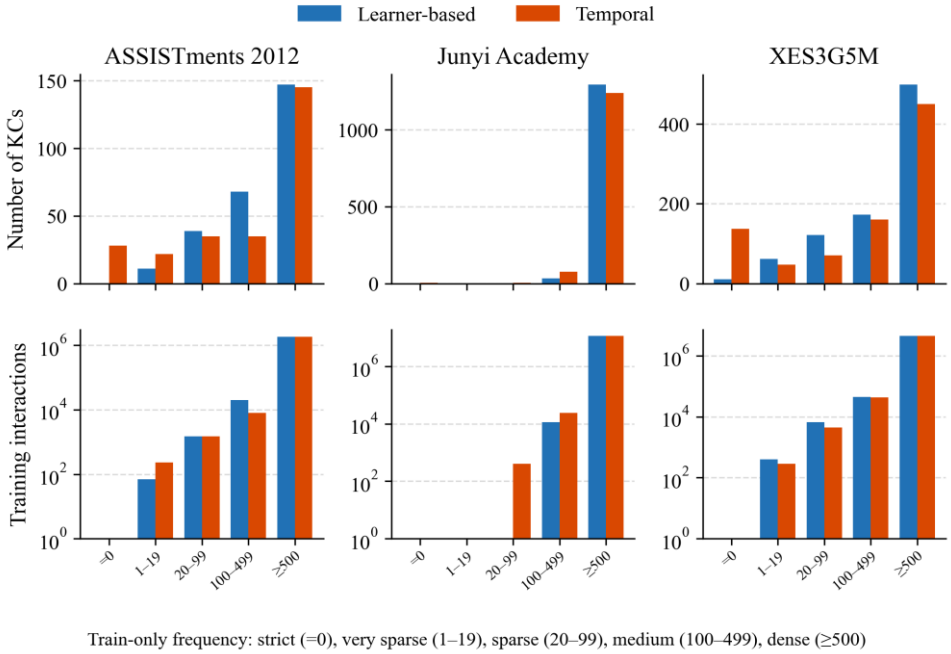


Fig. 2. Number of KCs (top) and number of training responses (bottom; sum of f_{train} in fold 0, logarithmic axis) in each training-frequency group under the learner-based and temporal splits.

D. Notation, frequency groups, and sample support

Each row of a processed log is a tuple (u, q, c, y). Learner u answered item q, c is the KC attached to the row, and $y \in \{0, 1\}$ records whether the response was correct. A partition divides the learners into disjoint training, validation, and test sets, and D_{test} denotes the test rows with a valid exported prediction. The training frequency $f_{train}(c)$ of KC c is its number of training rows. A group map g assigns each KC to one of five groups: strict cold start ($f_{train} = 0$), very sparse ($0 < f_{train} < 20$), sparse ($20 \le f_{train} < 100$), medium ($100 \le f_{train} < 500$), or dense ($f_{train} \ge 500$). Fig. 2 shows how many KCs and training responses fall in each group. The test rows of group k are

$$T_k = \{i \in D_{test} : g(c_i) = k\}, \quad N_k = |T_k| \tag{1}$$

and every group-level metric is computed on T_k alone. N_k is counted in each partition; the result tables report its mean over the four partitions and assign support labels from this mean, whereas the leakage audit (Section III.F) labels each partition. Analyses that use another subset, such as the threshold analysis, state their own N. A group is labeled Reliable (R) if $N \ge 1{,}000$, Limited (L) if $100 \le N < 1{,}000$, and Insufficient (I) if $N < 100$. These cut points are conventions of this study that flag sample support; they are neither confidence intervals nor hypothesis tests. A metric can be computed for any nonempty group but is interpreted only for Reliable or Limited groups; Insufficient groups are reported descriptively. Likewise, a sparse-versus-dense contrast is computable whenever both groups are nonempty, and we treat it as estimable if and only if (E1) some KCs have $f_{train}(c) < 100$ in the training split and (E2) the sparse and the dense group are both at least Limited; E2 presupposes E1. In each partition, the protocol thus (i) assigns groups from training data, (ii) labels their support, (iii) checks E1 and E2, (iv) reports discrimination and calibration by group, with the bias bound of Section III.E, and (v) may apply one fixed threshold to estimate decision errors (Section III.G).

E. Calibration metrics

Within a group, let $p_i \in [0, 1]$ be the predicted probability of a correct response and y_i the observed outcome. With $M = 15$ equal-width bins, ECE is

$$\mathrm{ECE} = \Sigma_m \, (n_m / N) \, |acc_m - conf_m| \tag{2}$$

where N is the number of rows in the group, n_m the number in bin m, and acc_m and $conf_m$ the observed proportion correct and the mean predicted probability in that bin. The Brier score is

$$\mathrm{Brier} = (1/N) \, \Sigma_i \, (p_i - y_i)^2 \tag{3}$$

and its binned Murphy decomposition [19] is

$$\mathrm{Brier} \approx \mathrm{UNC} - \mathrm{RES} + \mathrm{REL} \tag{4}$$

with $\mathrm{UNC} = \bar{y}(1 - \bar{y})$, $\mathrm{RES} = (1/N) \, \Sigma_m \, n_m (acc_m - \bar{y})^2$, and $\mathrm{REL} = (1/N) \, \Sigma_m \, n_m (conf_m - acc_m)^2$, where $\bar{y}$ is the proportion

correct in the group. Because the components are computed on binned probabilities, they need not sum exactly to the Brier score. Lower ECE and REL indicate better calibration, and neither replaces AUC. The main calibration contrast is

$$\Delta\text{ECE} = \text{ECE}_{\text{sparse}} - \text{ECE}_{\text{dense}} \quad (5)$$

where each term is the mean over the four learner-based partitions. Partition-level standard deviations describe variation across partitions and are not confidence intervals for ΔECE. Binned ECE also depends on the number of bins and is biased upward when bins contain few responses [29], [30], so a small sparse group can by itself make ΔECE positive. To bound this bias, suppose the outcomes in a group were independent draws with the predicted probabilities. Then $acc_m - conf_m$ has mean zero and variance at most $1/(4n_m)$, so $\mathrm{E}|acc_m - conf_m| \le 1/(2n_m^{1/2})$, and the Cauchy–Schwarz inequality gives

$$\mathrm{E}[\text{ECE}] \le \text{ECE}_0(N) = \tfrac{1}{2}\,(M/N)^{1/2} \quad (6)$$

For $M = 15$, ECE_0 is 0.095 at $N = 415$ and 0.003 at $N = 523{,}971$. By the same argument, the expected ECE of any model exceeds its value at the bins' expected accuracies by at most ECE_0, so the conservative contrast

$$\Delta\text{ECE}^* = \text{ECE}_{\text{sparse}} - \text{ECE}_0(N_{\text{sparse}}) - \text{ECE}_{\text{dense}} \quad (7)$$

is positive only if the sparse–dense difference exceeds this upper bound on expected small-sample bias. ECE0 is computed from the focal N of that partition. Fold 2 and fold 3 are averaged first, and the reported ΔECE* is the mean of the four unique partitions. It is not the result of subtracting ECE0 once from the mean ΔECE. ΔECE* is not a confidence interval, not a hypothesis test, and not a p-value. The size-matched gap, and the learner-cluster interval computed on fold 0 only, are different quantities.

F. Leakage checks

The protocol defines eight leakage checks. L1–L7 follow the numbering of the archived per-partition audit records, which cover only these seven. L1: training, validation, and test learners are disjoint. L2: every fitted transform uses training rows only; the row filtering of Section III.A fits no parameters. L3: the KC map is the static mapping supplied by the platform. L4: groups are assigned by applying the fixed cut points to f_{train} only. L5: no calibration map is fitted on test data. L6: no checkpoint or setting is selected with test data. L7: the strict cold-start group is defined by $f_{\text{train}}(c) = 0$. L8: when one response is expanded into several KC rows, the outcome recorded at one row does not enter the model input used to train on or score another row of the same response.

Table 3 reports the outcomes for the four learner-based partitions: checks L1–L7 passed in every partition. The archived audit covers the three local models; simpleKT reads the same partition files and selects epochs on validation learners only (Section III.C), so L1, L2, and L6 also hold for it by construction. The KC map (L3) is the released skill_id for ASSISTments 2012, the released KC mapping for XES3G5M, and the operational ucid for Junyi Academy. Check L8 does not apply to ASSISTments 2012 and Junyi Academy, where each released row records one response with one KC and no row was expanded. For XES3G5M, it was not part of the archived audit, and our audit of the code found that it fails (Section III.A). Test data were used only to compute the reported metrics.

Table 3. Leakage checks over the four learner-based partitions (L1–L7 numbered as in the archived audit records of the local models). PASS means that the condition held in every partition; parentheses give the occupancy of the affected group (I, Insufficient). L8 was audited from the code (Section III.A).

Check	Condition	ASSISTments 2012	Junyi Academy	XES3G5M
L1 Split	Training, validation, and test learners disjoint	PASS	PASS	PASS
L2 Preprocessing	Fitted transforms use training rows only	PASS	PASS	PASS
L3 KC map	Static, platform-provided mapping	PASS	PASS	PASS
L4 Frequency groups	Cut points applied to training counts only	PASS	PASS (sparse group empty)	PASS
L5 Calibration	No calibration map fitted on test data	PASS	PASS	PASS
L6 Model selection	No checkpoint or setting selected with test data	PASS	PASS	PASS
L7 Cold-start group	$f_{\text{train}} = 0$, defined from training data	PASS (empty in 3 partitions, I in 1)	PASS (empty)	PASS (I)
L8 Multi-KC rows	Outcome of a response hidden from the input for its other KC rows	N/A (one row per response)	N/A (one row per response)	FAIL (visible in training and test)

G. Secondary analyses

Two secondary analyses complement the group-level results. The first concerns KC structure. For each KC, we computed four training-only descriptors. The difficulty proxy is 1 − mean training correctness, an observed rate and not an IRT parameter. Item support is the number of distinct items per KC, learner exposure the number of distinct training learners per KC, and the curriculum-position proxy the median normalized position of the KC in learners' sequences. We report Spearman correlations of $\log(1 + f_{\text{train}})$ with the difficulty proxy (frequency–difficulty coupling) and with the curriculum-position proxy. A test-response-weighted regression relates T-KT's KC-level ECE to $\log(1 + f_{\text{train}})$ and the four descriptors, and a within-KC sparsification check reduces the training data of originally dense KCs.

The second is a simulated threshold analysis on the saved predictions. With one fixed threshold $\tau = 0.7$ for all groups, the simulated system advances the learner (skips remediation) when $p \ge \tau$ and assigns remediation otherwise. The false-advance rate is

$$\text{FAR} = P(y = 0 \mid p \ge \tau) \quad (8)$$

that is, the proportion of advance decisions followed by an incorrect response; FAR equals one minus the precision of the advance decision, or its false discovery rate. We also report three related quantities: the false-positive rate $\text{FPR} = P(p \ge \tau \mid y = 0)$; the expected FAR implied by the model's own probabilities, $\mathrm{E}[1 - p \mid p \ge \tau]$; and the excess FAR, observed minus expected. The contrast $\Delta\text{FAR} = \text{FAR}_{\text{sparse}} -$

FAR_{dense} is positive when advances on sparse KCs fail more often. FAR in (8) is computed over the N_{adv} responses with $p \geq \tau$, and FPR over the N_{inc} incorrect responses. Where prediction-level files exist, 95% intervals come from a KC-clustered percentile bootstrap (B = 2,000, with KCs resampled within the sparse and dense groups). Because y records the next response and not latent mastery, FAR is a simulated proxy for decision error.

IV. RESULTS

A. Overall discrimination

Table 4 reports overall AUC and accuracy on rows with a nonempty test window. Under the primary protocol, ASSISTments DKT has AUC 0.7303 ± 0.0005 and accuracy 0.7354 ± 0.0008. T-KT masked has AUC 0.7244 ± 0.0007 and accuracy 0.7314 ± 0.0006. The mean and SD use four unique partitions. Junyi DKT, under the same scoring rules but as a post-hoc check, has AUC 0.7554 ± 0.0012. simpleKT on ASSISTments, also post-hoc, has AUC 0.7711 ± 0.0013 and is not ranked against the primary models. These figures are not benchmark scores from a different split.

Table 4. Overall AUC and accuracy on nonempty test windows (mean ± SD across four unique learner partitions). N is that mean count. Fold 2 and fold 3 are one partition and are averaged before the SD. The SD is descriptive. Protocol P0-BV200-15EW. Junyi DKT and simpleKT are post-hoc sensitivity rows, not primary claims.

Dataset	Model	AUC ± SD	ACC ± SD	N	Protocol ID
ASSISTments 2012	DKT (primary)	0.7303 ± 0.0005	0.7354 ± 0.0008	523611.5	P0-BV200-15EW
ASSISTments 2012	T-KT masked (primary)	0.7244 ± 0.0007	0.7314 ± 0.0006	523611.5	P0-BV200-15EW
Junyi Academy	DKT (post-hoc)	0.7554 ± 0.0012	0.7469 ± 0.0013	3204343.0	P0-BV200-15EW
ASSISTments 2012	simpleKT (sensitivity)	0.7711 ± 0.0013	0.7533 ± 0.0014	523611.5	P0-BV200-15EW
XES3G5M	Any model	Not evaluated	Not evaluated	Not evaluated	NONE

B. Calibration across frequency groups

On ASSISTments 2012, under protocol P0-BV200-15EW, sparse ECE is higher than dense ECE for both primary models (Table 5). For DKT the four-partition means are 0.0982 ± 0.0093 sparse and 0.0089 ± 0.0013 dense, so ΔECE = 0.0893 ± 0.0091. ΔECE* = −0.0081 ± 0.0122. The size-matched gap is 0.0400 ± 0.0107. On fold 0 that gap is 0.0351, and the learner-cluster 95% interval is 0.0170 to 0.0821 (N sparse = 430). ΔECE* on that fold is −0.0088. The matched gap is positive on all four partitions. ΔECE* is positive only on P1. The interval is for fold 0. It is not a test of the other partitions, and ΔECE* is not that interval. For T-KT masked the four-partition sparse and dense ECE are 0.1199 ± 0.0247 and 0.0158 ± 0.0032 (ΔECE = 0.1041 ± 0.0215). ΔECE* = 0.0068 ± 0.0207, positive on P1 and P4 and negative on P0 and P2. The matched gap is 0.0581 ± 0.0230. The fold-0 gap is 0.0365, with interval 0.0233 to 0.0878. The mask, the checkpoint, and the window were introduced together, so this is not a mask-only effect. simpleKT is sensitivity only and is not in Table 5.

Junyi Academy has no sparse group. Its DKT medium-versus-dense contrast uses the same checkpoint and window, but it is post-hoc. The four-partition means are ECE 0.0345 ± 0.0080 on medium and 0.0043 ± 0.0021 on dense (ΔECE = 0.0302 ± 0.0069; mean N medium = 3676.75). ΔECE* = −0.0032 ± 0.0067. That value is the mean of the per-partition references, not 0.0302 minus the bound at a single N. The matched gap is 0.0115 ± 0.0073. The fold-0 interval is 0.0010 to 0.0434 and applies to that partition. XES3G5M has no model row. Check L8 failed, the windows repeat responses at some boundaries, and no corrected model was trained. Its frequency distribution remains a descriptive audit.

The base-rate reference shows why ECE has to be read together with resolution. On dense ASSISTments KCs its ECE was only 0.0031, yet its resolution was zero (Table A1): a nearly constant predictor can appear well calibrated while carrying no ranking information. Because its predictions fall in one bin, its ECE equals the gap between that prediction and the observed proportion correct; on sparse KCs this gap (0.0777) exceeds the one-bin version of (6), 0.025, so it reflects a lower proportion correct rather than small-sample bias. The reference is thus neither a lower bound on attainable ECE nor a competitive model.

Fig. 3 shows the fold-0 reliability diagrams for the two primary models on ASSISTments 2012. The sparse curves use 430 responses and 15 bins. They are visual checks, not the four-partition estimate.

Table 5. Calibration under protocol P0-BV200-15EW (mean ± SD across four unique partitions; 15 equal-width bins; nonempty windows). The primary models are ASSISTments DKT and T-KT masked. ΔECE* uses each partition's own N and is a bias reference, not a confidence interval or a test. The 95% interval is the fold-0 learner-cluster interval of the size-matched gap only; it is not a result for every partition. A higher sparse ECE is not attributed to the checkpoint alone. Support: Limited if 100 ≤ N < 1,000, Reliable if N ≥ 1,000. Junyi is post-hoc and has no sparse group. XES3G5M is a descriptive audit only and is not evaluated.

Dataset	Model	Group (support)	N	ECE	ΔECE	ΔECE*	Matched gap	95% CI (fold 0 gap)
ASSISTments	DKT	sparse (Limited)	397.75	0.0982 ± 0.0093	0.0893 ± 0.0091	−0.0081 ± 0.0122	0.0400 ± 0.0107	0.0170 to 0.0821
ASSISTments	DKT	dense (Reliable)	517329.5	0.0089 ± 0.0013	same contrast	same reference	same gap	same interval
ASSISTments	T-KT masked	sparse (Limited)	397.75	0.1199 ± 0.0247	0.1041 ± 0.0215	0.0068 ± 0.0207	0.0581 ± 0.0230	0.0233 to 0.0878
ASSISTments	T-KT masked	dense (Reliable)	517329.5	0.0158 ± 0.0032	same contrast	same reference	same gap	same interval

Dataset	Model	Group (support)	N	ECE	ΔECE	ΔECE*	Matched gap	95% CI (fold 0 gap)
Junyi (post-hoc)	DKT	medium (Reliable)	3676.75	0.0345 ± 0.0080	0.0302 ± 0.0069	−0.0032 ± 0.0067	0.0115 ± 0.0073	0.0010 to 0.0434
Junyi (post-hoc)	DKT	dense (Reliable)	3200666.3	0.0043 ± 0.0021	same contrast	same reference	same gap	same interval

C. Estimability and cold-start groups

Whether a sparse contrast can be estimated is a property of the counts (Table 6; RQ2). ASSISTments 2012 has a nonempty sparse group. Under the primary protocol its mean sparse N is 397.75, which is Limited, and its dense group is Reliable, so the contrast is eligible. Junyi Academy has no sparse group; the post-hoc contrast is medium versus dense, and the medium group is Reliable. XES3G5M has low-frequency KC rows, but the model contrast is not eligible: check L8 failed and the dataset is a descriptive audit only. Strict cold-start and very-sparse groups are not given a primary ECE.

The structural descriptors in Table 6 characterize the datasets, and none of them is treated as a necessary or sufficient condition for the direction of a contrast. Frequency and difficulty were negatively correlated on ASSISTments 2012 ($\rho = -0.227$, with sparse KCs harder) and Junyi Academy ($\rho = -0.416$) but weakly positively correlated on XES3G5M ($\rho = 0.110$).

Table 6. Conditions for estimating a sparse-versus-dense contrast, structural context, and calibration direction. E1 is computed on learner-based fold 0 over the KCs present in training; test-row counts are means over four learner-based partitions. Part B uses learner-based fold 0 (KCs with test rows; for XES3G5M, KCs present in training). Very sparse: $0 < f_{train} < 20$; strict cold start: $f_{train} = 0$. IQR, interquartile range. ΔECE* > 0 refers to (7); in the medium rows, the medium group takes the place of the sparse one. XES3G5M model-based cells are not reported (check L8 failed).

Question	ASSISTments 2012	Junyi Academy	XES3G5M
Sparse group present?	Yes	No	KC-level tail only; not a model contrast
L1–L7	PASS for the local models	PASS for the local models	Not a license for model scores
L8	Not applicable (one KC per row)	Not applicable (one KC per row)	FAIL
Eligible model contrast?	Yes: DKT and T-KT masked	Post-hoc medium versus dense only	No
Direction	Matched gap positive; ΔECE* mixed	Matched gap positive; ΔECE* −0.0032	Not evaluated
Uncertainty	Fold-0 matched CI; four-partition SD is descriptive	Fold-0 matched CI; post-hoc	Not evaluated
Protocol	P0-BV200-15EW primary	Same rules, not pre-specified	Descriptive audit only
Primary claim?	Yes	No	No

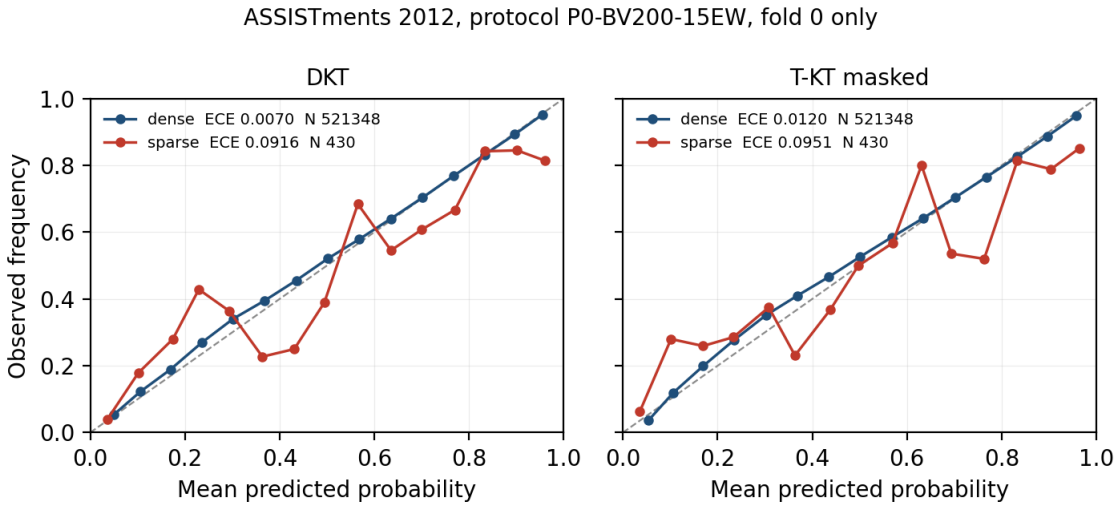


Fig. 3. Reliability diagrams for the primary protocol on ASSISTments 2012, fold 0, 15 equal-width bins, nonempty windows. Left: DKT. Right: T-KT masked. Dense and sparse groups only. N sparse = 430 in each panel. This is one partition, not a test of the other three. Overall AUC can hide a sparse-versus-dense calibration difference, but this figure does not show that a checkpoint removed or caused it. Table 5 holds the four-partition means. ΔECE* in that table is a bias reference, not a significance test. Junyi has no sparse group and is not plotted. XES3G5M is a descriptive audit only and is not plotted.

D. Robustness to frequency cut points

Table A2 is the ASSISTments DKT grid of checkpoint and test window. Every cell is ΔECE. No cell is missing. On fold 0, final × full is 0.1689 and final × window 200 is 0.1620, so the window alone moves sparse ECE only from 0.2251 to 0.2179. At that same window, the validation-AUC checkpoint moves sparse ECE from 0.2179 to 0.0916 (ΔECE 0.1620 to 0.0846). Final × full versus validation × window 200 changes both factors; that gap is not assigned to the checkpoint alone and is not assigned to the window alone. Raw ΔECE stays positive in every cell. ΔECE* does not. The alternative frequency grids of the archived final-checkpoint analysis were not repeated here and are not a current result.

E. KC-level associations and controlled sparsification

A test-response-weighted regression of T-KT's KC-level ECE on $\log(1 + f_{\text{train}})$ and the four descriptors compares different KCs in learner-based fold 0. The standardized frequency coefficient (95% interval, KC-clustered standard errors) was negative on ASSISTments 2012 (−0.063 [−0.081, −0.045]; 260 KCs) and Junyi Academy (−0.006 [−0.009, −0.003]; 1,326 KCs). Difficulty (positive) and curriculum position (negative) were also associated with ECE on both datasets, and item support and learner exposure on ASSISTments 2012. Because T-KT was trained and scored under different contexts (Section III.C), these associations describe that implementation only.

Reducing the training data of originally dense KCs did not produce a uniform direction. The check covered 12 cells (ASSISTments 2012 and Junyi Academy, DKT and T-KT, and reductions to 500, 100, and 50 training rows; 30 originally dense KCs each; seed 42). Three cells, all for T-KT on Junyi Academy, had 95% KC-bootstrap percentile intervals (B = 10,000) entirely above zero, four had intervals entirely below zero, and five included zero; no DKT cell had an interval entirely above zero. The per-cell grid is in the artifact. The group contrast, the regression, and this check estimate different quantities, so their signs need not agree.

F. Threshold analysis

A threshold probe of false advances was not recomputed on the primary-protocol predictions. FAR is Not evaluated under P0-BV200-15EW. Tables A3 and A4 remain an archived final-checkpoint probe. They are not a finding of this protocol, and they are not used in the abstract or the conclusion.

V. DISCUSSION

A. What the results show

The results support a limited claim. Overall AUC can hide frequency-stratified calibration differences, but the size and the sign of the extra summaries depend on the protocol, the checkpoint, the sample size, the model, and the dataset. On ASSISTments 2012, DKT and T-KT masked have a positive raw ΔECE and a positive size-matched gap. The fold-0 interval for that gap excludes 0. ΔECE* does not exclude a small-sample explanation on every partition, and it is not a significance test. The fold-0 interval is not evidence that all four partitions are significant. Junyi Academy has no sparse group. Its post-hoc ΔECE* is −0.0032. XES3G5M is a descriptive audit only. The study does not claim that a validation checkpoint removes sparse miscalibration, and it does not claim that the final checkpoint causes it.

B. Why the datasets differ

The datasets differ first in whether a sparse contrast exists (Table 6). Junyi Academy's exercise-level components all have at least 100 training responses under this split, so the sparse group is empty. ASSISTments 2012 has a Limited sparse group and is the only dataset with a primary model contrast. XES3G5M can describe frequency and estimability, but its model scores were produced by a route that fails L8. Those scores are excluded. The models differ as well. simpleKT was not a primary comparison. The study does not identify a mechanism that would make rare practice itself the cause of miscalibration.

C. Implications for adaptive learning platforms

For platforms that use KT probabilities to skip practice, assign remediation, or declare mastery, the results point to four checks before a single global threshold is trusted. (a) Occupancy: count KCs and test responses by training frequency in the split actually used, and report empty or low-support groups. (b) Calibration: report ECE and the Brier decomposition by group alongside AUC, and compare small-group ECE with the bias bound (6). (c) Decision error: if a fixed threshold is evaluated, report FAR and FPR together with their denominators. (d) Leakage and alignment: verify the checks in Table 3 in training as well as in scoring; when a response carries several KCs, construct the model input so that no KC row of the response reveals its outcome to another; in attention-based models, mask the target and later responses during training; and match each exported prediction to the right learner, item, KC, and sequence position.

Post-hoc calibration methods such as temperature scaling [17] and Platt scaling [31] were outside our scope; whether they close the gap on sparse KCs remains open.

D. Limitations

Several limits qualify these findings. The label is the next response, not a mastery decision, and no classroom trial was run. The primary contrast is one dataset, ASSISTments 2012, with a Limited sparse group. Four unique partitions describe variability. They are not four independent population samples, and fold 2 is the same learners as fold 3. ΔECE* is a bias reference. The matched interval covers fold 0 only. T-KT masked changed the mask, the checkpoint, and the window together. The archived full-history pairs are not this protocol, and their weights were not recovered. Junyi DKT is post-hoc. Junyi T-KT was not retrained with a mask. XES3G5M is a descriptive audit only. BKT was not scored. FAR was not recomputed. None of the analyses identifies a causal effect of training frequency on calibration.

VI. CONCLUSION

Overall AUC can hide frequency-stratified calibration differences, but their size and presence depend on the evaluation protocol, the checkpoint, the sample size, the model, and the dataset. On ASSISTments 2012, under the locked validation-AUC and window-200 protocol, DKT and a causally masked Transformer show a higher sparse ECE than dense ECE. The fold-0 size-matched interval excludes 0. The four-partition bias reference ΔECE* is −0.0081 for DKT and 0.0068 for the masked Transformer, and that reference is not a significance test. Junyi Academy has no sparse group; its post-hoc ΔECE* is −0.0032. XES3G5M is a descriptive audit only, with no model-based result. The study does not recommend a new tracing model or a classroom policy. It asks evaluations that feed instructional rules to report calibration and support by training frequency, and to keep the

bias reference separate from a size-matched interval.

APPENDIX

Table A1. Archived final-checkpoint, full-history Brier decomposition. Not a result under protocol P0-BV200-15EW. Not used as a current finding.

Dataset	Model	Group	N	Flag	Brier	UNC	REL	RES
ASSISTments 2012	Base rate	dense	523,971	R	0.2103±0.0003	0.2103±0.0003	0.0000±0.0000	0.0000
ASSISTments 2012	Base rate	medium	5,963	R	0.2742±0.0075	0.2484±0.0013	0.0258±0.0062	0.0000
ASSISTments 2012	Base rate	sparse	415	L	0.2420±0.0218	0.2337±0.0129	0.0084±0.0090	0.0000
ASSISTments 2012	DKT	dense	523,971	R	0.1931±0.0010	0.2103±0.0003	0.0053±0.0005	0.0223±0.0004
ASSISTments 2012	DKT	medium	5,963	R	0.2297±0.0144	0.2484±0.0013	0.0301±0.0086	0.0479±0.0053
ASSISTments 2012	DKT	sparse	415	L	0.2502±0.0098	0.2337±0.0129	0.0624±0.0038	0.0451±0.0037
ASSISTments 2012	T-KT	dense	523,971	R	0.2116±0.0019	0.2103±0.0003	0.0201±0.0015	0.0186±0.0004
ASSISTments 2012	T-KT	medium	5,963	R	0.2272±0.0086	0.2484±0.0013	0.0273±0.0021	0.0478±0.0053
ASSISTments 2012	T-KT	sparse	415	L	0.2525±0.0116	0.2337±0.0129	0.0594±0.0104	0.0392±0.0088
Junyi Academy	Base rate	dense	3,226,541	R	0.2078±0.0007	0.2078±0.0007	0.0000±0.0000	0.0000
Junyi Academy	Base rate	medium	3,706	R	0.2579±0.0033	0.2432±0.0013	0.0147±0.0021	0.0000
Junyi Academy	DKT	dense	3,226,541	R	0.1804±0.0008	0.2078±0.0007	0.0022±0.0002	0.0294±0.0002
Junyi Academy	DKT	medium	3,706	R	0.3077±0.0144	0.2432±0.0013	0.0741±0.0123	0.0091±0.0033
Junyi Academy	T-KT	dense	3,226,541	R	0.1892±0.0034	0.2078±0.0007	0.0091±0.0022	0.0275±0.0009
Junyi Academy	T-KT	medium	3,706	R	0.2347±0.0074	0.2432±0.0013	0.0163±0.0040	0.0251±0.0045

Table A2. ASSISTments DKT sensitivity grid. Each main cell is ΔECE = ECE sparse − ECE dense. best/200 is the primary protocol. ΔECE* in the extra column is the bias reference for that primary cell, not a confidence interval. Fold 2 and fold 3 share learners. No cell is missing. This table does not attribute the final/full versus best/200 gap to one factor.

Fold	final/full	final/200	best/full	best/200	ΔECE* best/200	N sparse best/200	reason missing
fold 0	0.1689	0.1620	0.0977	0.0846	-0.0088	430.0000	none
fold 1	0.1746	0.1732	0.1017	0.1016	0.0074	423.0000	none
fold 2	0.1086	0.1084	0.0905	0.0867	-0.0118	386.0000	none
fold 3	0.1354	0.1316	0.0867	0.0935	-0.0050	386.0000	none
fold 4	0.1319	0.1335	0.0822	0.0808	-0.0224	352.0000	none

Table A3. Archived threshold probe. Not evaluated under protocol P0-BV200-15EW. Not a current finding.

Model	Group	N	N_{adv}	N_{inc}	FAR [95% CI]	E[FAR]	Excess	FPR
T-KT	dense	528,018	284,326	158,623	0.196 [0.186–0.208]	0.113	0.083	0.352
T-KT	sparse	444	235	197	0.268 [0.202–0.337]	0.050	0.218	0.320
DKT	dense	528,018	315,650	158,623	0.200 [0.190–0.211]	0.147	0.053	0.398
DKT	sparse	444	243	197	0.296 [0.221–0.383]	0.057	0.239	0.365

Table A4. Archived threshold probe across partitions. Not evaluated under protocol P0-BV200-15EW. Not a current finding.

Model	Mean ΔFAR	SD	Range	Partitions with ΔFAR > 0	Mean N	Mean N_{adv}	Mean N_{inc}
T-KT	0.056	0.031	0.015 to 0.087	4/4	413	227	155
DKT	0.040	0.052	−0.011 to 0.096	3/4	413	226	155

CONFLICT OF INTEREST

The authors declare no conflict of interest.

AUTHOR CONTRIBUTIONS

Author 1: Conceptualization, Methodology, Software, Formal analysis, and Writing (led diagnostic design, experiments, and manuscript). Authors 2 and 3: Software (data processing and baseline runs). Author 4: Methodology (methodological review). Author 5: Supervision and Writing (manuscript revision). All authors approved the final version.

ETHICAL STATEMENT

This study is a secondary analysis of three publicly released educational datasets (ASSISTments 2012, Junyi Academy, and XES3G5M) that the providers de-identified before release. We did not recruit participants, contact learners, parents, guardians, or teachers, or conduct any classroom intervention, and we collected no new consent forms, logs, recordings, or personal data. No direct identifiers were accessed, and no attempt was made to re-identify any learner. The data were used under the providers' terms; Junyi Academy releases its data under CC BY-NC-SA 4.0, and XES3G5M is released under the MIT license. The study was neither commissioned by nor conducted with an educational technology company. Under the research policy of the authors' institution, a secondary analysis of publicly released, de-identified data does not require prospective human-subjects review, so no ethics protocol number applies. No exemption from data-protection law is claimed.

DATA AND CODE AVAILABILITY

The analysis artifact contains configuration files, preprocessing, training, and audit scripts, the saved summaries behind the reported tables, per-partition audit records, and scripts that, without retraining, rebuild selected diagnostic tables and recompute the bias bound (6) and the count-based rows of Table 6 from these files. It excludes the raw datasets, available from their providers, the prediction files, and the code of the controlled sparsification check, which was not archived; retraining requires those datasets and may not reproduce the reported values exactly. An anonymized copy is available to reviewers through the editor,

and a public repository will be released upon acceptance.

GENERATIVE AI STATEMENT

During manuscript preparation, the authors used Cursor 3.18.9 (with Grok 4.6 and Grok 4.7) for language polishing, formatting, consistency checking, and preparation of reproducibility prompts; ChatGPT (OpenAI; GPT-5.6) to edit an intermediate revision and to compare revised versions; and Claude (Anthropic; Claude Opus 5.5) to review the manuscript and the analysis code and to draft structural and language revisions. No AI tool was used to generate or alter experimental results. The authors checked every AI-assisted passage and code finding against the underlying code and results, revised it, and take full responsibility for the content. No AI tool is listed as an author.

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